

# ORC Waste Heat to Power Market Size, Share and Forecast 2035

The ORC waste heat to power market is emerging as a vital component of global energy efficiency and decarbonization strategies. The market size was over USD 5.4 billion in 2025 and is projected to reach USD 13.9 billion by the end of 2035, expanding at a compound annual growth rate (CAGR) of 11.1% during the forecast period from 2026 to 2035. Strong growth is supported by rising industrial energy consumption, tightening emission regulations, and increasing focus on recovering unused thermal energy from industrial and natural sources.

**Request Sample@** <https://www.researchnester.com/sample-request-5570>

## ORC Waste Heat to Power Industry Demand

[ORC waste heat to power](#) systems convert low- to medium-temperature heat into electricity using organic working fluids with low boiling points. Unlike conventional steam-based systems, ORC technology enables efficient power generation from waste heat streams that would otherwise be released into the environment.

Industry demand for ORC systems is driven by their high operational efficiency, cost-effective power generation, and ease of integration with existing industrial processes. These systems require limited supervision, offer automated operation, and are designed for long service life with low maintenance needs. Their ability to operate across diverse heat sources—including industrial exhaust, geothermal reservoirs, and biomass combustion—has positioned ORC systems as a practical solution for improving energy utilization and reducing operational costs.

## ORC Waste Heat to Power Market: Growth Drivers & Key Restraint

### Growth Drivers

- **Rising Focus on Energy Efficiency and Emission Reduction**  
Governments and industries are increasingly prioritizing waste heat recovery to improve energy efficiency and meet sustainability goals. ORC systems enable power generation without additional fuel consumption, making them attractive for emission-intensive industries.
- **Technological Advancements in ORC Systems**  
Continuous improvements in turbine design, working fluids, digital monitoring, and system modularity have enhanced performance, reliability, and scalability. These advancements are expanding adoption across both industrial and renewable energy applications.
- **Cost-Effectiveness and Operational Flexibility**  
ORC systems offer favorable lifecycle economics due to reduced fuel costs, minimal

water usage, and stable long-term operation. Their ability to function efficiently under variable heat loads further supports industrial deployment.

### **Restraint**

- **High Initial Capital Investment**

Despite long-term cost benefits, upfront installation costs and project-specific engineering requirements can limit adoption, particularly among small and mid-sized industrial facilities.

## **ORC Waste Heat to Power Market: Segment Analysis**

### **Segment Analysis by Technology**

Organic Rankine Cycle systems dominate the market due to their adaptability across a wide temperature range and proven industrial reliability. Kalina cycle systems are used in specialized applications where variable heat conditions exist, while thermoelectric generators are gaining interest for small-scale and decentralized waste heat recovery.

### **Segment Analysis by Heat Source**

Industrial processes remain the primary heat source, particularly in energy-intensive manufacturing sectors. Geothermal energy applications support stable, base-load power generation, while biomass-based systems enable combined energy recovery and waste reduction. ORC, Kalina, and thermoelectric technologies are increasingly integrated depending on heat quality and operational requirements.

### **Segment Analysis by Application**

Power generation represents the core application, supporting both grid-connected and captive power needs. District heating systems benefit from ORC integration for improved energy utilization, while combined heat and power configurations enhance overall system efficiency in industrial and municipal settings.

### **Segment Analysis by Capacity**

Small-scale systems are widely used in decentralized and modular installations, while medium-capacity systems support industrial facilities with consistent waste heat output. Larger installations are typically associated with heavy industries and renewable energy projects requiring continuous power generation.

### **Segment Analysis by End-User Industry**

Cement and steel manufacturing leads adoption due to high-temperature exhaust availability. Petrochemical and chemical industries deploy ORC systems to improve plant efficiency, while oil and gas operations use waste heat recovery to offset power demand in remote or energy-intensive environments.

### **Segment Analysis by Cycle Type**

Subcritical ORC systems are preferred for their simplicity and operational stability across standard industrial applications. Supercritical ORC systems are gaining traction in advanced installations due to their higher efficiency potential and suitability for higher heat-source temperatures.

### **ORC Waste Heat to Power Market: Regional Insights**

#### **North America**

The North American market is driven by strong industrial efficiency initiatives, supportive energy policies, and growing investments in waste heat recovery across manufacturing and oil & gas sectors.

#### **Europe**

Europe represents a mature and innovation-driven market, supported by stringent emission regulations, widespread adoption of CHP systems, and strong focus on renewable integration and circular energy use.

#### **Asia-Pacific (APAC)**

APAC is the fastest-growing region, fueled by rapid industrialization, expanding manufacturing capacity, and increasing emphasis on energy efficiency in emerging economies. Large-scale infrastructure and industrial projects continue to drive strong demand.

### **Top Players in the ORC Waste Heat to Power Market**

The ORC waste heat to power market is moderately consolidated and includes prominent players such as Turboden (Italy), Ormat Technologies (U.S.), Exergy International (Italy), ElectraTherm (U.S.), Enertime (France), Orcan Energy (Germany), Enogia (France), Rank (Spain), Triogen (Netherlands), Zuccato Energia (Italy), Mitsubishi Heavy Industries (Japan), IHI Corporation (Japan), Kawasaki Heavy Industries (Japan), Kaishan (China), Kaishan USA (U.S.), Echogen Power Systems (U.S.), Siemens Energy (Germany), General Electric (U.S.), Thermax (India), and Doosan Enerbility (South Korea), all actively focusing on efficiency enhancement, system scalability, and global market expansion.

**Access Detailed Report@** <https://www.researchnester.com/reports/orc-waste-heat-to-power-market/5570>

**Contact for more Info:**

**AJ Daniel**

**Email:** [info@researchnester.com](mailto:info@researchnester.com)

**U.S. Phone:** +1 646 586 9123

**U.K. Phone:** +44 203 608 5919